Stage 5 — Machine-Learning Survival Models

Questions 1–3 | XGBoost, prediction horizons, and leaderboard comparison

Q1. Why didn’t the trees win?

The paired centre-grouped cross-validation comparison (G – D) gives a mean IPCW C-index difference of **+0.0020**, standard deviation **0.0035**, with XGBoost winning **11 of 15 folds**. The tuning favoured relatively small trees (depths 2–4 and roughly 100–200 trees); the final model used depth 4 and 100 trees. In the feature-importance plot, **ipssr_score contributes 19.4%** of total gain.

Trees can capture nonlinear effects and interactions, but engineered predictors such as ipssr_score and cyto_ipssr_cat already encode combinations of clinical and cytogenetic information. The preference for shallow trees suggests limited additional interaction signal. With approximately 1,600 observed deaths, fitting more complex trees also risks overfitting. As a result, XGBoost performs only slightly better than elastic-net Cox (approximately 0.724 versus 0.722 in CV), while the random survival forest is lower (approximately 0.719).

Q2. Reading the horizon plot

Model	At 1 year	At 7 years	Drop
Elastic-net Cox	0.7562	0.7249	0.0313
IPSS-R	0.7148	0.6738	0.0410

Elastic-net drop: 0.7562 – 0.7249 = **0.0313**.
IPSS-R drop: 0.7148 – 0.6738 = **0.0410**.

Elastic-net advantage at 1 year: 0.7562 – 0.7148 = **0.0414**.
Elastic-net advantage at 7 years: 0.7249 – 0.6738 = **0.0511**.

Early deaths often occur among patients with pronounced high-risk features at diagnosis. Seven-year outcomes are harder to rank because treatment, relapse, and later disease changes are not fully represented by baseline data. The elastic-net advantage over IPSS-R grows at seven years, suggesting added clinical and molecular features help with longer-term discrimination. Late observed deaths also receive larger IPCW weights: the paper gives about 1.46 at one year versus 25.61 at seven years.

Q3. Do the leaderboard and cross-validation agree?

Model	CV	Public	Public – CV
Elastic-net Cox	0.722	0.7543	+0.0323
IPSS-M-style Cox	0.716	0.7374	+0.0214

Elastic-net: 0.7543 – 0.722 = **+0.0323**.
IPSS-M-style: 0.7374 – 0.716 = **+0.0214**.

Both public scores exceed their centre-grouped CV means. The unseen KYW test centre has a higher-risk case mix, including more TP53 mutations and complex karyotypes and lower haemoglobin and platelet counts. That may make some risk differences easier to rank, but censoring patterns and the public subset may also affect the score; the case mix alone does not prove the explanation.

> **XGBoost public-score prediction:** Its CV improvement over elastic-net is 0.7238 – 0.7218 = **0.0020**. Applying the same increment to the elastic-net public score gives 0.7543 + 0.0020 = **0.7563**. This is a rough estimate, not a guaranteed result: the CV improvement is small relative to its variability, and the actual XGBoost public score was not provided.

Conclusion. The results suggest that careful feature engineering already captures much of the available prognostic structure. XGBoost offers a small possible improvement, while elastic-net Cox remains a strong, more interpretable comparator.